Use environmental information and riding data to predict bicycle gear positions

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***Abstract*—The main purpose of this thesis is to obtain the physiological information of the rider, the current environmental information, and select a suitable gear so that the rider can ride smoothly without worrying about the problem of gear switching. We use the previously established smart bicycle system as a prototype to collect the physiological state of the rider and the current environmental state in real time. Several different decision-making methods are used with the collected sensor information to predict and analyze the current suitable gear. Due to the use of deep learning methods, a certain amount of data is required for training. We use the previously established bicycle system collect the physiological information and environmental data of the rider. After slightly adjusting the system, we use the system to collect the current multiple sensor information of the rider, as well as the current gear status. These data are used to train the machine learning method we have established, and finally methods are used for testing. In addition, we have established a web platform that can be used to manage the collected data. The platform can be used for data upload, query, and simple analysis for viewing.**

I. Introduction

The hardware part of this paper is to use the smart bicycle system established by [1] as a prototype, and adjust it for use. The system is a platform that can collect the physiological state of the rider and the current environmental state in real time. The system at that time also analyzed and predicted the collected data, but because it focused on the establishment of the platform, it only used the random forest method to predict the gear.

In recent years, there have been more and more methods based on machine learning as a means of decision-making, but in the field of bicycles, most data sets only contain information such as the starting point and ending point and the corresponding time. If only these information are obviously predicting our gear Training is not very effective. So we collect a data set ourselves, which contains the physiological information of the rider, the current environment information, and the current gear information, so that we can use machine learning methods for training.

Figure 1 is a system flow chart based on [1]. The system is divided into two parts, RaspberryPi and Arduino. Arduino is mainly used as a platform for collecting data and controlling gears. After collecting the data, it is transmitted to RaspberryPi through I2C. RaspberryPi then collects GPS data (position and speed). The mentioned decision-making method determines a suitable gear state based on the information and then sends it back to the Arduino to switch the gear.

II. Related Work

In the published literature, the decision-making methods of bicycle speed change are as follows: the frequency of pedaling and the method of adding fuzzy theory and the method of using the relationship between pedal frequency and wheel speed to make decisions, there is also a method to estimate the road gradient based on the pressure of the pedals and the speed of the vehicle, and then combine the load of the vehicle to determine the appropriate power of the electric vehicle [2]. And through the speed of the vehicle and the rotation speed of the crank to determine the decision of the electric bicycle to upshift and downshift. These methods use the speed of the car, crank speed and other methods to estimate the decision to change gears, or the motor power, but these methods lack physiological information for the rider, and for the slope of the real environment, and the road. Analysis of the situation.

In the two papers [3] [4], it is known that when the rider is in a calm state, the speed of the heart rhythm is directly related to the frequency of exercise. In [5] it is mentioned that although many competitive cyclists ride at a lower pedaling frequency in a climbing environment, the research [6] shows that a higher pedaling rhythm can be improved when going uphill. Riding performance. And this article uses the IMU provided by [7] combined with the Kalman filter to estimate the tilt state of the car body and filter out the noise, which is used as a part of the reference basis for adjusting the gear.

In the part of gear decision we used Random Forests [8], Deep Forest [9], XGBoost [10], LightGBM [11], CATBoost [12], and because Recently, the method of using neural networks for decision-making is quite popular, and even [13] specially builds the table data into a picture to realize the neural network of the table data, so we also have the method of using the neural network To make gear decisions.

Using the above methods to collect information such as IMU, heart rhythm, foot pressure, temperature information, GPS, etc., by using pre-collected information as input and manually changing gears as ground truth training models

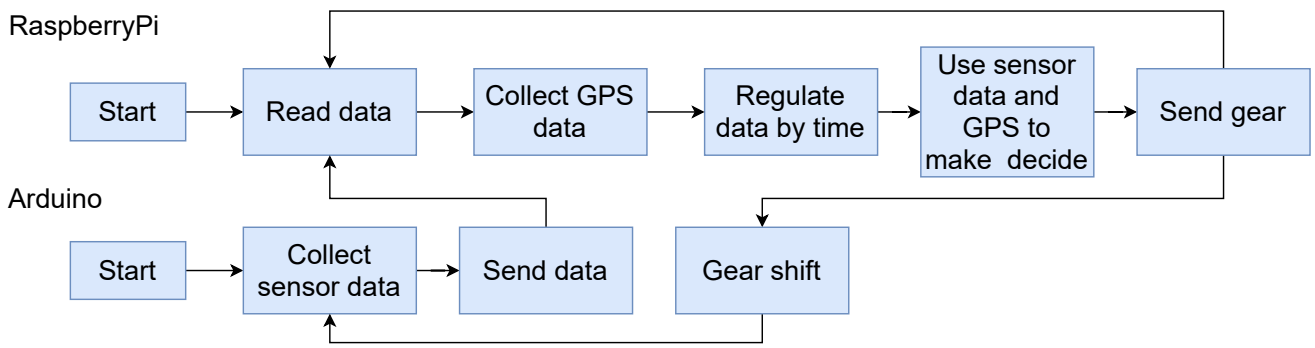


Fig. 1. Simple flow chart

for real-time detection Use physiological and environmental information to make gear decisions.

III. Hardware system

This section will first review the architecture of the hardware platform [1] used, and explain the adjustments and additions. Next, the operation process of this system will be introduced in addition to the part of gear prediction, including how to collect data when collecting training data and when actually testing.

A. Hardware review

This subsection introduces and explains the system of [1], as well as the additions and adjustments in this paper. Picture 2 is the overall picture of the system. Figure 2The part in red box 5 is the core of the whole system, including the Arduino and RaspberryPi are installed in that position, and the RTC time module, GPS module, IMU module, and electronic transmission system shift buttons The switch is also installed on this plane, and the Arduino serves as the main sensing information collection and motor control. RaspberryPi is used as a data analysis platform and can upload data to the database when necessary. The electronic transmission system of this system is driven by a motor as shown in the red box 1 of 2, and the red box 4 of 2 is the power source of the system. Next, the red box 2 of 2 is the pressure sensor of the system. Figure 2Red box 7, this part is a finger pulse sensor. Figure 2 The red frame 6 is a Hall effect sensor, which is used with a magnetic strip fixed on the inner side of the wheel to estimate the riding speed.

B. Hardware adjustment

First of all, we believe that the temperature information will also affect the riding state of the rider, so we added a temperature sensor module to the system to detect the ambient temperature. In this paper, the NTC thermistor temperature sensor, Heat-Sensitive Temperature Switch Sensor Module w is selected.

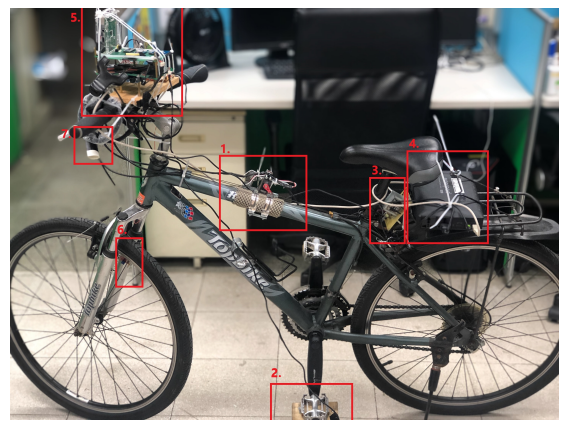


Fig. 2. Physical image of bicycle

The speed sensing part was originally realized by using a Hall-effect sensor with a magnet fixed on the inner side of the wheel. However, Hall effect sensors need to be intensively sampled when the wheel is rotating. During this period, they cannot be distracted to sample other sensors, and the Arduino is a single-core platform, so it is impossible to use a multi-thread approach to stand alone. The core performs sampling, which takes up too much calculation of the entire Arduion, reduces the sampling frequency of the overall sensor, and a window period occurs, which makes the sampling rate of the entire system unsmooth, resulting in low credibility of the collected data. As an alternative, the speed information provided by GNSS is used as an alternative when collecting training data.

C. System process and architecture

Figure 3 is the flow chart of the system when collecting data, because there is no need to use sensor data for gear judgment, so there is no part of transmitting Arduino data to RaspberryPi, and the part of switching gears. Replace it with a button, and use this gear message as the ground turth, and also remove the part that repeats the confirmation of the network status. In addition, the sensor sampling part chooses to use protothread to schedule the sampling time of each sensor, so that the sampling frequency can be controlled more easily, and the scheduling frequency of the IMU can also be increased to replace high-frequency interrupts.

Figure 3 At the beginning of the process, Arduino will first initialize the sensor and motor, set the interrupt subroutine for shifting, use protothread to set the sampling sequence and frequency of each sensor, and then borrow The set schedule keeps the system running, and during the period when the interrupt is triggered, it decides to increase or decrease according to the button. RaspberryPi does not need to analyze the sensor data of Arduino, so the main work is only to collect GPS information, including speed and location information. The GPS information will be processed by RaspberryPi mainly because GPS connection takes time and RaspberryPi is more stable than Arduino Many, and can be controlled wirelessly through SSH and other methods.

The process of the test phase is because the test phase needs to use sensor data, GPS (including speed and location information), and map data to make timely gear judgments during riding. So the main difference between this part and the picture 3 is that the Arduino data is transmitted to RaspberryPi through I2C, and the part where Arduino originally used buttons to manually shift gears is replaced by waiting for the RaspberryPi to determine the gear position, and then start the gear. Bit switching.

IV. Gear analysis system and web page construction

This section first introduces and explains the gear prediction method used in this paper. The gear prediction method we use includes Random Forests, Deep Forest, XGBoost, LightGBM, CATBoost, and neural networks to make decisions. Then we will explain and introduce the functions of the cloud database system and the website that we have built.

A. Data analysis and gear forecast

For the method of gear prediction, we chose to test several methods including Deep Forest, XGBoost, LightGBM, CATBoost, Random Forests, and neural networks.

The input feature information includes pedal pressure, heart rate, IMU, ambient temperature, GPS, and moving speed. The above is some basic information. In addition, map information and past gear are added. Use these two information to combine basic sensor information. Used together for gear prediction. The past gear information is added because only real-time information is used to make a gear decision. The data will be in the middle of the two gears, resulting in frequent changes in the gear during this time period. This way of shifting gears does not conform to the riding habits of the public. Therefore, we add past gear to the judgment, so that it can learn how to switch gears based on the previous gear and sensor data, and learn that gear changes will not happen frequently in a short time.

When XGBoost, LightGBM, CATBoost, Deep Forest, Random Forests directly predict our data, the accuracy is not low, but there are high-frequency gear changes. If the information of past gear is directly added, it will be excessively dependent on the gear information. Therefore, we use a two-stage prediction method for training. As shown in the figure 4, we will first use the input information without past gear to make a prediction, and the results will have high frequency changes. If this information is used As a past gear, because it is not a real ground turth, there will be no over-reliance. Therefore, we will add the predicted gears as the information of the past gear to the input characteristics at the previous time point in order to proceed. Training once, then use the information containing the past gear to train in the second stage of training, repeat the second stage a certain number of times and then stop to get the final result. In this way, the past gear can be considered while not being overly dependent on the past gear.

In the part of the neural network, the prediction method we choose is mainly to refer to the Artificial Neural Network [14], and also to refer to the part of the model architecture of ResNet [15] and UNet [15].

In addition, we have added information from the map to analyze some artificial scene information. In artificial scenes, there are often some obvious slopes that have no obvious changes, and there are no obvious changes in physiology. However, it is indeed necessary to change gears, such as It is the traffic lights, intersections, etc., such as these situations, using the information of the map to make a judgment in advance. It can effectively enable riders to have a better riding experience.

B. Cloud building and web page hosting

The web page is built on the basis of the web page set up by [1]. The database uses the MySQL platform. The web page template is provided by HTML5 UP. [1], the main functions include administrator login, riding information query, data registration, data analysis, and the administrator can query the status of multiple riders.

The structure is shown in Figure 5, which is mainly divided into two parts before login and after login. Before login, only the homepage information can be viewed. Other related functions need to be logged in before browsing operations, such as data upload and data Query, gear analysis and other functions.

V. Data collection

The dataset can be roughly divided into two parts: inside and outside the school. Because the information in the school is

[1]HTML5 UP web page template license: https://html5up.net/ license

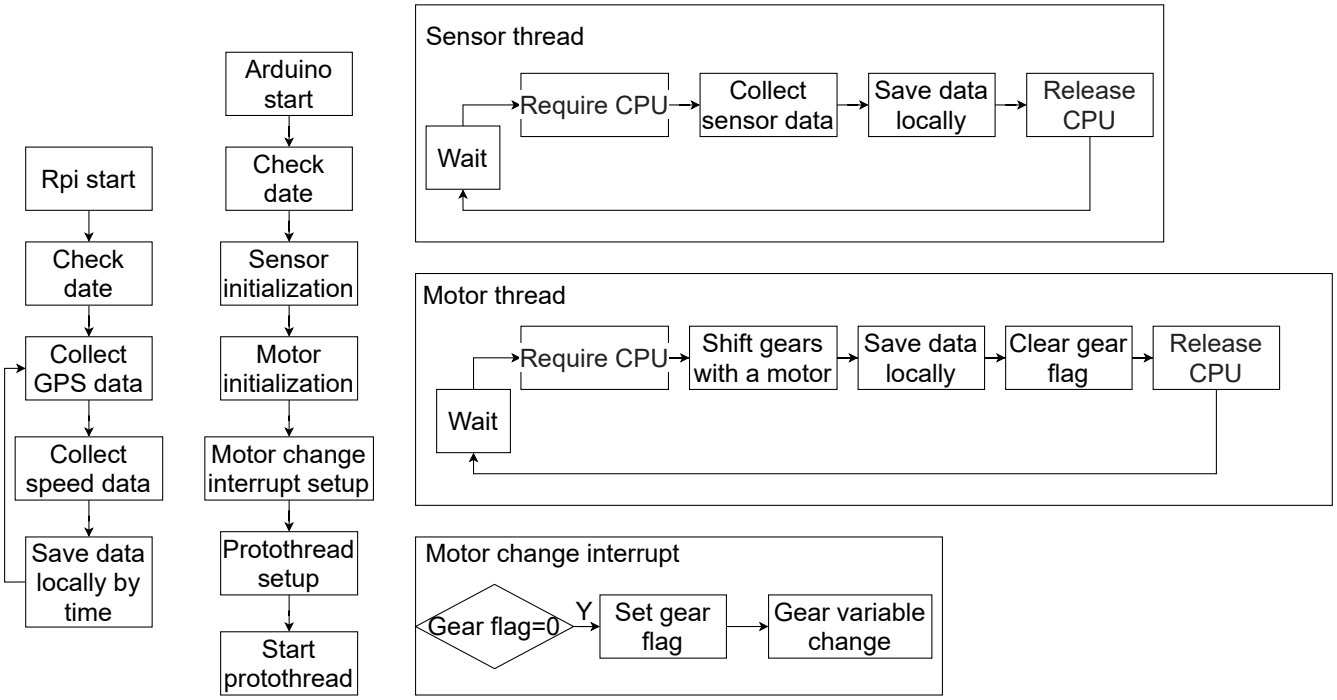


Fig. 3. Flow chart of collecting training data

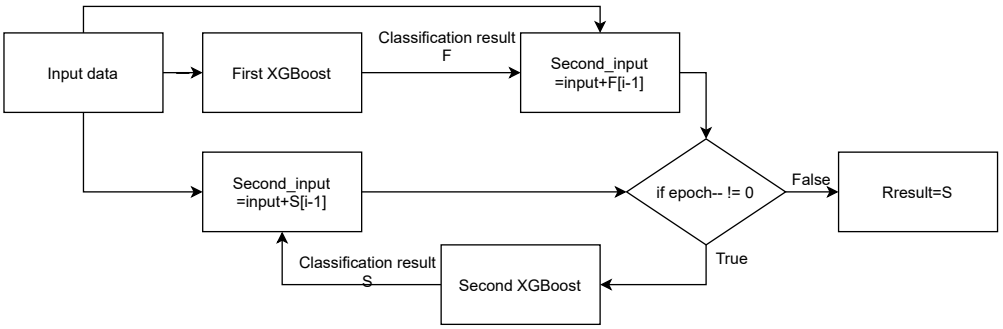


Fig. 4. Two-stage training: the first stage of training does not consider the past gear, the second stage uses the first-stage prediction result as the past gear information for training, and repeats epoch times

in a more standardized state, there will be a fixed and planned route for riding when collected. Because the outside of the school is relatively open, although there is still a slight plan for the route when riding, it will not completely follow the pre-planned action, and the riding habits are slightly different from the state of the school.

Figure 6[ae] is the main route for collecting data in the school, and because it is a more standardized route, roughly 10,000 data are collected for each route, and a total of 50,000 data are collected for training. And each route has different road conditions including uphill, downhill, and flat ground, and there are two routes that contain important road markings.

The picture 6[f] is the status of all the routes included in the data set outside the school we have collected. The total amount of data is about 50,000 for training.

VI. Experiment

In the testing part, we will be divided into several chapters to test different methods separately. The picture 7 is the data set we use to conduct school tests, and its scope covers many scenes that appear in the training data.

A score evaluation formula 1 suitable for our application situation is designed. The standard is set as the ratio of accuracy minus the variation score osc and the number of samples sa . Because we don't want the gear to switch too frequently, we set the change score to negative. The method

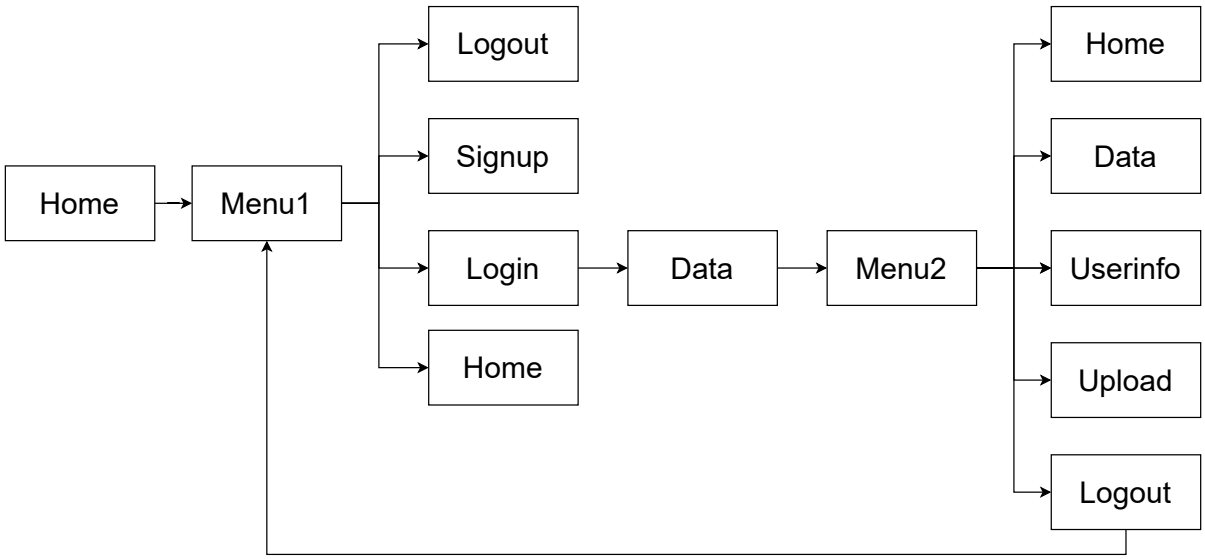


Fig. 5. Web page structure diagram, data query and upload can only be performed after login, and only login and registration related operations can be performed before login

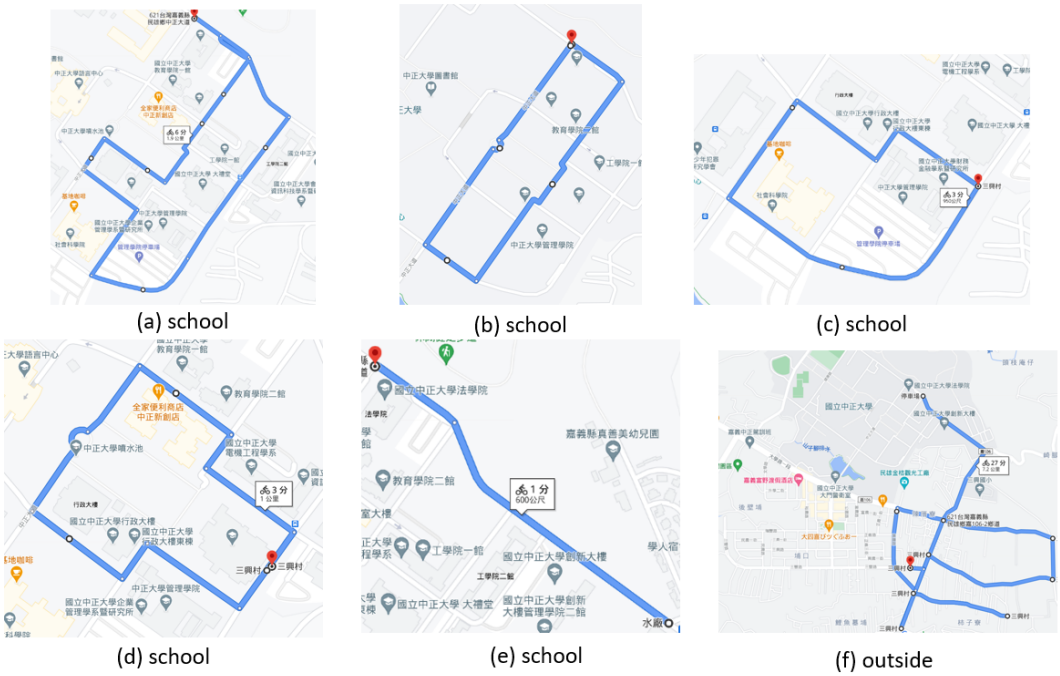


Fig. 6. Training dataset

is to use a sliding window with a size of sl to scan from the beginning to the end of the test data. When the gear is within 2 seconds G changes more than twice, accumulate the number of changes $\times$ and divide by the number of samples. In addition, the accuracy is compared by the ground turth and the prediction result. When the two are the same, the count is counted and the overall ratio is calculated.

$$\begin{aligned} Score &= Accuracy - osc/sa \\ osc &= \sum_{n=0}^{n=sa-sl} \begin{cases} 0 & \mu < 2 \\ \mu & \mu \geq 2 \end{cases} \\ \mu &= \sum_{m=n}^{m=n+sl-1} \begin{cases} 0 & G[m] = G[m+1] \\ 1 & G[m] \neq G[m+1] \end{cases} \end{aligned} \tag{1}$$

A. Neural network

First, let's discuss the method of using neural networks. This part of the test data contains information about important markers. The red box in the picture 7[b] is when the marker is passed, it is promoted early because of the marker. For the gear, you can see that the picture 7[b] predicted this.

In order to confirm the impact of adding past gear to our entire architecture, there is a special comparison between considering past gear and not considering past gear's respective forecast conditions under the same architecture.The left picture of the picture 8 does not consider the condition of the past gear. It can be seen that there are many frequent oscillations in the whole picture, and the picture on the right of the picture 8 is the condition of the past gear. It is obvious that compared to the left picture without considering the past gear, those high-frequency oscillations have disappeared. From these two pictures, it is obvious that adding past gear to the network for judgment can be more in line with human riding habits. .

B. Random Forests

We use the Random Forests method used in the [1] paper for testing. The left image of the picture 9 is the effect of using the original Random Forests. It can be seen that the overall profile and accuracy are not bad, but the frequency of gear shifting is too high overall. This should be because the past gear is not considered The relationship of making judgments. Therefore, in the right picture of the picture 9, we added the information of the past gear to make a decision. We can see that the overall gear change frequency has been reduced a lot.

C. Deep Forest

Use DeepForest to analyze our data. The left picture of the picture 10 uses the original DeepForest method and does not consider the previous gear for testing. It can be seen that the overall frequency of gear changes is too high. In line with the riding habits of the human body, the picture on the right of the picture 10 has added the information of the previous gear. It is obvious that the frequency of gear changes is reduced.

D. XGBoost,LightGBM,CATBoost

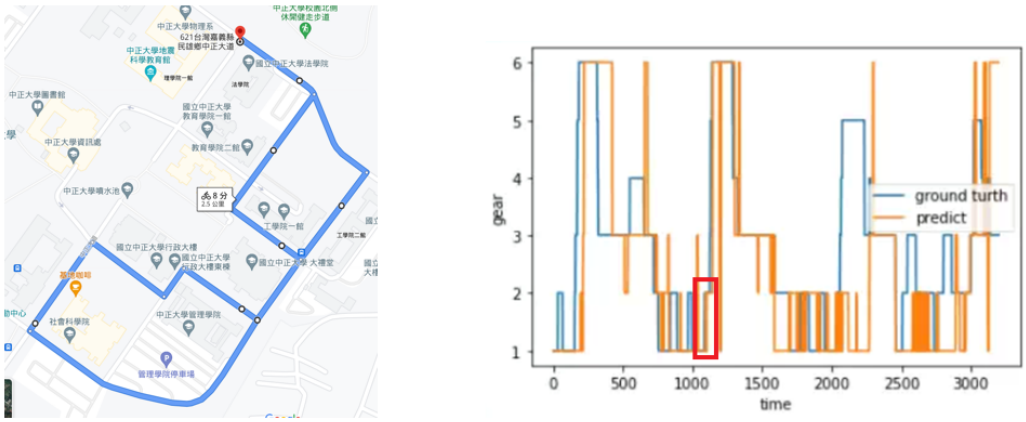
Finally, we use XGBoost, LightGBM, CATBoost these three methods to analyze our data set, the left picture of the picture 11 is the test result after using the original XGBoost to adjust the parameters, you can see that the XGBoost method itself has The accuracy is inaccurate, and the overall prediction profile is also in line with the ground turth. However, because the past gear information is not taken into account during the entire prediction process, the overall prediction results often change frequently. In order to improve this Situation We have added the two-stage training method mentioned in the third and added the message of past gear in the training. The result is shown in the right figure of 11, and the overall stability can be seen from the figure. It has improved a lot, there is almost no problem of frequently changing gears, and the accuracy is not much different from the original XGBoost. The left picture of 12 13 also shows the LightGBM and CATBoost without considering the previous gear. The accuracy of the two methods has a certain level, but the frequency of gear changes is still too high. Picture 12 13 The picture on the right shows the use of a two-stage training method after adding the previous gear, the overall stability has improved a lot.

It can be seen from the above experiments that although many methods can have good accuracy in terms of accuracy without adding past gear, there will always be a problem of too high gear switching frequency. So we used the method of adding past gear to training. Use the dilution or two-stage training method to add the past gear information to the training. It can also be seen that the overall fluctuation frequency of XGBoost, RF, LightGBM, and CATBoost using the two-stage method is lower than that of purely diluted NN, DF, and RF. This shows that our two-stage training method is effective.

E. Discussion

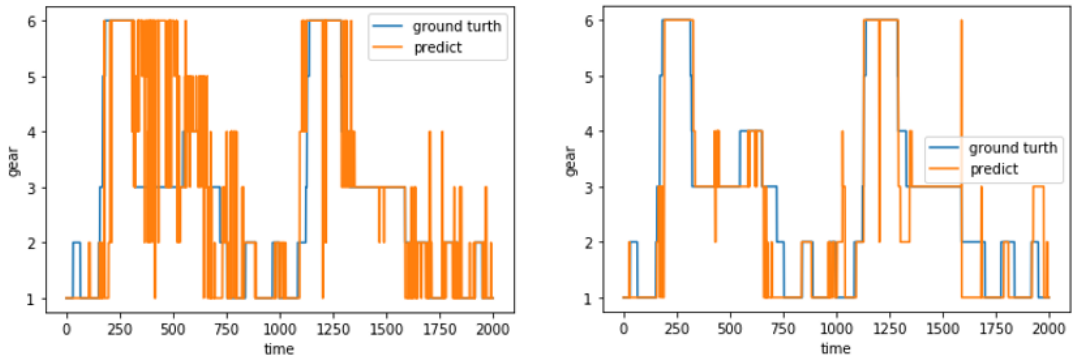
From the previous experiments, it can be known that among the neural network, Random Forests, Deep Forest, XGBoost, LightGBM, and CATBoost, the neural network method does not perform as expected in the table data set. The remaining few In this method, without considering the previous gear, the accuracy difference of the several methods is not large. After adding the previous gear, XGBoost, LightGBM, CATBoost, Random Forests benefit from the possibility of multiple training in an iterative manner. Obtained more satisfactory results than DeepForest, of which XGBoos, CATBoost these two methods have the best effect. After all the materials are added to the training, because the riding habits of the school and off-school materials are different, the effect is a little less than that of the training alone, or only slightly improved.

Table I Table II is the accuracy of all methods tested, the horizontal axis is the method of use, the vertical axis is the test dataset, and ALL represents the use of school and outside training together. It can be seen from the table that XGBoost and CATBoost have the highest scores, so we think Boosting Tree is very effective for our dataset. Although LightGBM is also the Boosting Tree method, but because



(a) Test path (b) Test result

Fig. 7. School test



Ignore the gear of the past Consider the gear of the past

Fig. 8. NN compares whether past gear is considered

of its lightweight structure, and LightGBM [11] mentioned that the data tested is far from our data, the dataset and the number of features mentioned in the paper The size of is much larger than ours, so we think this method is not suitable for our data. RandomForests still has good accuracy, but it is slightly inferior to XGBoost and CATBoost. The DeepForest effect using deep random forests is even worse than RandomForests, so we believe that our data does not require a very deep structure to effectively predict. Finally, the neural network method is still not suitable for our tabular dataset. Compared with the neural network, the tree structure method can obviously produce better results.

VII. Conclusion

This paper is based on the collection of physiological information and environmental information of bicycle riders established by [1], coupled with an electronic transmission system, to adjust the sampling frequency and system stability of the system, and use the adjusted platform We collected training data on 5 different riding routes in the school and off-campus routes, and collected a piece of on-campus and off-campus data as our test route. Use several machine learning

TABLE I
Score summary table

	XGB	CATBoost	LGBM	RF
school	0.785	0.808	0.732	0.727
ALLschool	0.687	0.800	0.756	0.746
outside	0.669	0.515	0.361	0.642
ALLoutside	0.44	0.558	0.389	0.688

TABLE II
Score summary table

	DF	NN
school	0.685	0.61
ALLschool	0.65	0.586
outside	0.456	0.478
ALLoutside	0.396	0.481

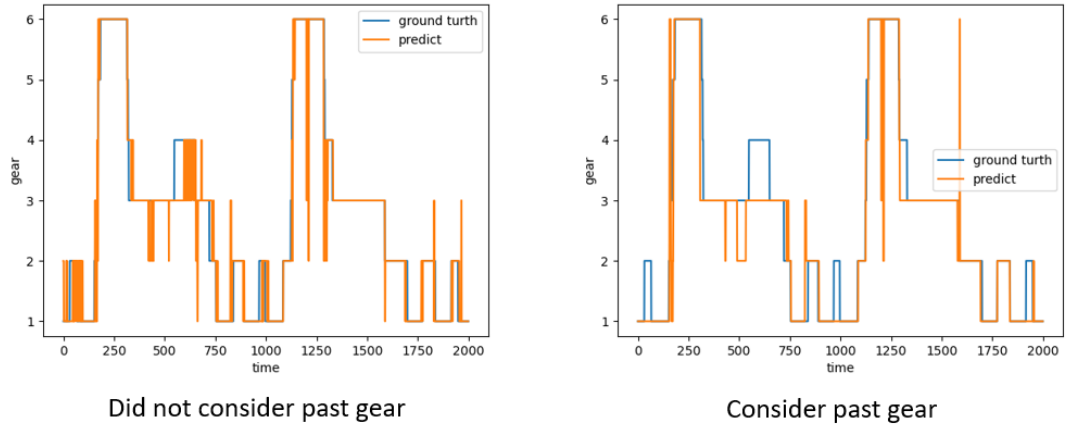


Fig. 9. RF compares whether past gear is considered

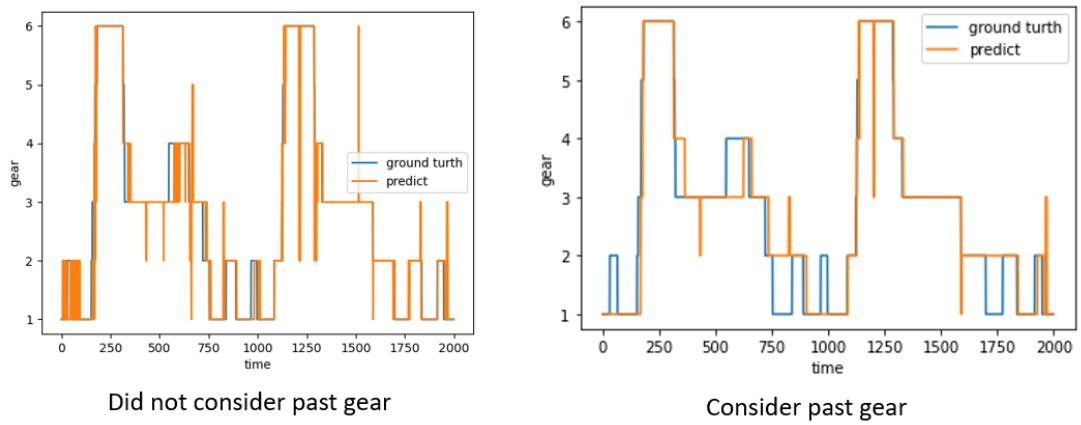


Fig. 10. DF compares whether past gear is considered

and deep learning methods to predict the position of the vehicle on this data set. The methods used include neural networks, RandomForests, DeepForest, XGBoost, LightGBM, CATBoost, and these methods are adjusted. Optimize, analyze and discuss all the results. Finally, we believe that the neural network method is still unable to effectively predict the table data. Among the other methods, RandomForest, XGBoost, LightGBM, and CATBoost benefit from repeated learning. The most satisfactory results are achieved, among which the predictions of XGBoost and CATBoost are the smoothest and conform to the riding habits of the test data. In addition, we have established a web platform for data storage, query, and simple analysis and comparison. For users to understand their riding conditions, and to confirm whether the system's predictions are consistent with the real conditions.

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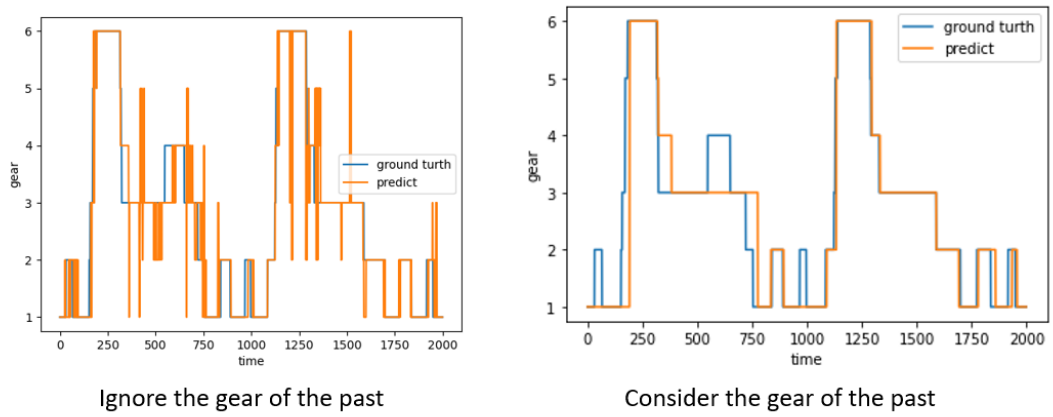

Fig. 11. XGBoost compares whether past gear is considered

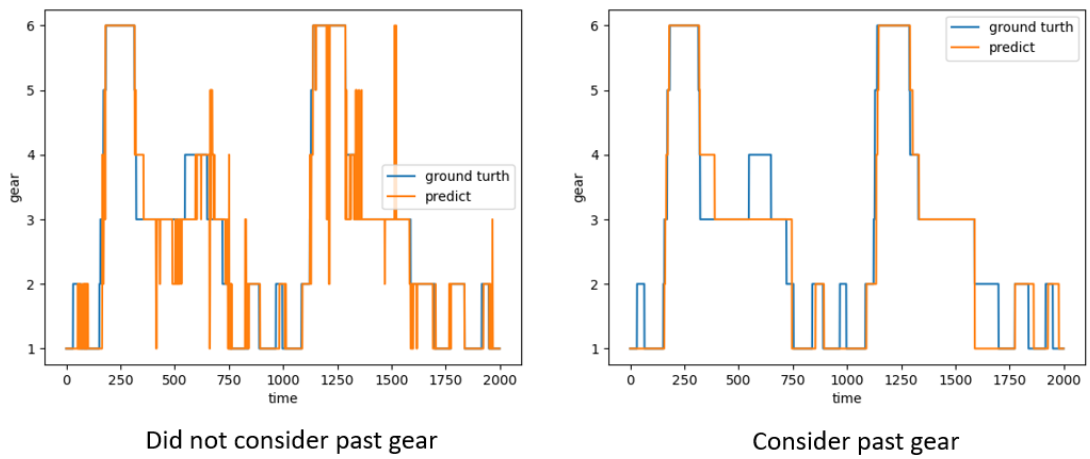

Fig. 12. LightGBM compares whether past gear is considered

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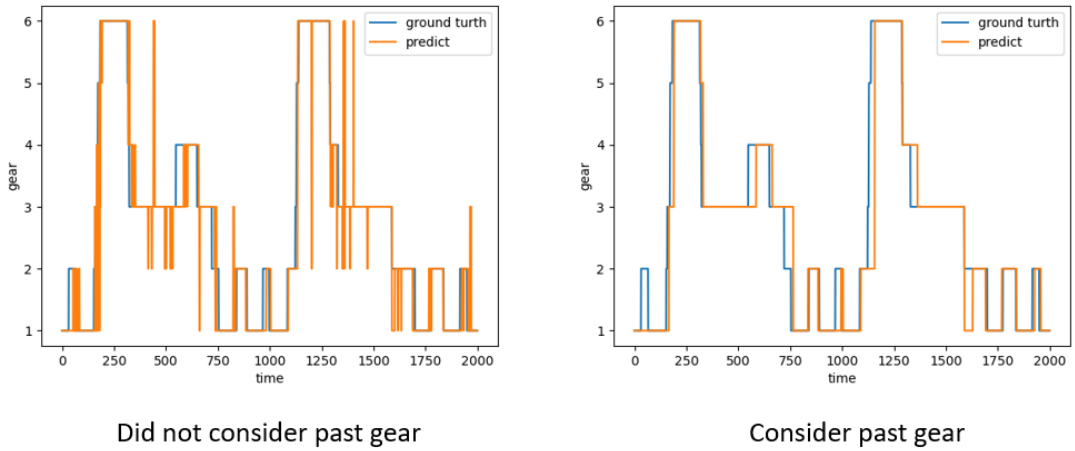

Did not consider past gear

Consider past gear

Fig. 13. CATBoost compares whether past gear is considered